1. Team Purpose or Mission

What does your team hope to accomplish by the end of the project? What questions will your dashboard help users explore, and who is the intended audience? How will your combined skills and interests support the work? What can your audience expect in terms of analytical quality, clarity, and professionalism?

- Think through personal goals in terms of skill building and keep it in mind throughout the project
    - Elizabeth: general R skill-building
    - Ashlee: build R skills
    - James: R statistics skill-building
    - Trisha: general competence in R, getting raw data comprehensible to a broad audience
- Dashboard/storymap intended audience: general audience, forest managers, urban planning potentially, large forest land owners
    - Questions to explore:
        - Fast fire presence/movement and its relationship with ecosystem boundaries/land cover species composition
        - Should I plan to change the species composition of the forest I own?
- How will our combined skills and interests support the work?
    - Ashlee brings great skills in communication and writing
    - Trisha brings experience from two previous R classes and is comfortable with the data-cleaning parts of this process
    - James has experience working with spatial data and can support bringing the spatial data in an R-usable framework
    - Elizabeth bringing a MF background established an ecological standpoint and helps framing how the deliverables can be used/ how they are important
- What can your audience expect in terms of analytical quality, clarity, and professionalism?
    - Hopefully high but tbd
    - Keep in mind that we don’t want to bite off more than we can chew

2. Communication Plan and Collaborative Coding

Which channels will you use for routine communication, urgent questions, and sharing files or code? How quickly should teammates respond? When are members generally available to work, and when will you meet regularly? How will you document decisions and next steps so everyone stays informed?

- Which channels will you use for routine communication, urgent questions, and sharing files or code
    - Text is great
- How quickly should teammates respond?
    - Ideally all team members can respond within 24 hours, but as soon as possible.
    - If the message is informing the group that someone is in the repository, liking the message to acknowledge is helpful
- When are members generally available to work, and when will you meet regularly?
    - We will meet every week on Thursdays during class time.
    - 4:00-4:30 Tuesday meeting. Aim to have most of our work for the week done by this meeting time.
- How will you document decisions and next steps so everyone stays informed?
    - Notes document in the project folder

3. Roles and Responsibilities

What interests each member about the project? What strengths do you bring, and what skills do you want to develop?

- Ashlee is interested in how fire intensity and activity is changing and how those changes impact ecosystems. She brings great skills in communication and writing. She hopes to generally further develop R skills since this is her first experience in using this coding language.
- Trisha is interested in understanding how landscape features can affect the rapid spread of wildfires. She brings experience from two previous R classes and is comfortable with the data-cleaning parts of this process. She hopes to generally improve her competence in R and in getting raw data comprehensible to a broad audience.

- James has experience working with spatial data and can support bringing the spatial data in an R-usable framework. He hopes to build deeper skills in statistical analysis in R as well as getting generally more comfortable with the format.
- Elizabeth is interested in understanding how different trees and forest communities may be more or less resilient to fast fires. She brings a MF background and an established ecological standpoint that can help frame how the deliverables can be used/ how they are important. She hopes to further develop her R skills generally.

How will you divide responsibilities for finding and preparing data, exploratory analysis, dashboard development, interpretation, documentation, and presentation?

- We will split up responsibilities for each individual milestone so that we all get the same practice in but the GitHub horrors are limited.
    - It is too early at the moment to say what exactly this will look like because we will want to assess each person's interests and skill development goals as the project progresses.
- We have the goal of sharing the workload evenly throughout the course.

How will you ensure that everyone contributes meaningfully and understands the complete project?

- Each Thursday, we will discuss the division of work for the coming week to ensure that everyone is contributing equally and meaningfully. We also want to be sure to support each other, explaining our work to others to improve all of our understanding.

Identify who will review each other's work and how responsibilities may change as the project develops.

- We will establish a lead and review role for each task. As we go along, we will ensure that the division of responsibilities align with each team member's interests and that they are divided equally amongst all team members.

4. Operating Norms or Ground Rules

How will you prepare for and run meetings? How will you assign tasks, set internal deadlines, and track progress? What should teammates expect when someone completes work independently—for example, organized files, readable code, and documented data-cleaning decisions? What will you do if someone anticipates missing a meeting or deadline? How will you combine and review work before submitting it?

How will you prepare for and run meetings?

- We will meet Tuesday afternoons 4-4:30pm, and aim to have each of our tasks mostly complete by that time. The Tuesday meeting serves as time to synthesize and either turn in or get very close to turning in our deliverable for that week.
- Our Thursday meetings during class time will be for coworking and for establishing the plan and distribution of work for the following Tuesday's deliverable.

How will you assign tasks, set internal deadlines, and track progress?

- We will use our Running Meeting Notes document in our shared google drive to keep track of action items and internal deadlines. As we complete our assigned tasks we will check them off in that document.

What should teammates expect when someone completes work independently—for example, organized files, readable code, and documented data-cleaning decisions?

- Teammates can expect all of the above– organized files, readable code, and documented data-cleaning decisions.
- When commits are made, the work done will be clearly described in the commit comment.
- Team members will send the team a message when they start working on a file in the repository and once they have pushed the file and stopped work.

What will you do if someone anticipates missing a meeting or deadline?

- Team members will do their best to notify the group as far in advance as possible if they will be unable to make a meeting or complete a task.
- If a team member is unable to make a meeting, they will review the notes taken in the Running Meeting Notes document. The remaining team members will share any action items that they've assigned to the team member who was unable to make it.

- If a team member is unable to meet a deadline they notify the remaining team members as soon as they can. That team member will provide detailed information so that the remaining team members can pick up that task in time for the deadline.
- If there is a week where one team member is very busy with other responsibilities, they will inform the group of that at their Thursday meetings, and plan to pick up more of the load for the group project in the following week so that overall work distribution is generally equitable.

How will you combine and review work before submitting it?

- Tuesday meetings will be for combining and reviewing work before submitting it. All team members will attend and participate in this synthesis.
- Team members will alternate who submits the deliverable.

5. Values and Behaviors

What does being a good teammate mean to your group? Agree on expectations for reliability, respectful communication, constructive feedback, and openness to different ideas.

- Team members should be timely and comprehensive with getting their items on the checklist we make done
    - Also in responding to communications
- Asking for help/clarification regarding confusion with the above step is encouraged
- Feedback and different ideas are both items that should be discussed during the Thursday meeting before we figure out what the next week of work looks like

How will you create space for all members to contribute and ask for help?

- The group chat is a great step!
- Some time during the Tuesday meeting to discuss what the trickiest part of each person's assigned task could go far in establishing the space to help the team members feel confident in asking for help

How will you address uneven participation or concerns about the quality of someone's work?

- The Tuesday meeting is a good time to address participation concerns
- Team members should be proactive about raising weeks where they may be super busy so that the teammates can support them

6. Resources and Constraints

What resources can support your work, including course materials, dataset documentation, instructors, TAs, and relevant subject-matter sources? What constraints should shape your scope—for example, the six-week timeline, missing data, differences in geographic or temporal coverage, technical skills, or other commitments?

Since follow-up with project partners may not be available, how will you resolve unanswered questions, document assumptions, and communicate limitations in the dashboard?

What resources can support your work, including course materials, dataset documentation, instructors, TAs, and relevant subject-matter sources?

- Posting the weekly deliverables clearly and ahead of time
- The desired level of dataset documentation should be clearly posted as well
- Instructors should be available enough to clear data sources/subject matter sources so that we can hit the ground running

What constraints should shape your scope—for example, the six-week timeline, missing data, differences in geographic or temporal coverage, technical skills, or other commitments?

- The six week timeline is pretty central in determining the project scope–we should make sure we don't bite off more than we can chew
- Geographic/temporal coverage issues/missing data should be documented, especially if it gets to the point of limiting our project scope

Since follow-up with project partners may not be available, how will you resolve unanswered questions, document assumptions, and communicate limitations in the dashboard?

- The README is a natural place to put these limitations
- In addition, they should be clearly spelled out in their relevant sections
- The Thursday discussion periods are a great time for individual team members to spell out concerns regarding questions, assumptions, and limitations to make sure they are documented

7. Decision-Making Guidelines

How will you make decisions about research questions, datasets, analytical methods, and dashboard design? How will you ensure that relevant teammates are involved and that choices are supported by evidence? What will you do if you cannot reach consensus? When should you seek guidance from an instructor or TA?

- Decisions about research questions, datasets, analytical methods, and dashboard design will be made as a group, guaranteeing that each member will have a say moving forward in the project. Each decision making step will be discussed either at our Tuesday afternoon meetings or while we are working during the Thursday class time. Making sure that each team member can bring their individual thoughts can solidify that each of our specific interests and goals for the project are met.
- To ensure that the relevant teammates are involved in the decision making, those that have prior experience regarding a certain section of manipulating datasets, analytical methods etc. will be encouraged to bring their expertise and suggest a viable option for our project. Bringing our unique backgrounds can give more routes the project can take, creating a more robust look at the dataset and the visualizations that can come from it.
- After sharing our potentially diverse viewpoints on what decision is best for the project, if we cannot reach a consensus on the particular section we can attempt to take both ideas into consideration. For example, if we were stuck on which statistical/regression model to use we could always do both and then compare them by their AIC or associated MSE etc. Also, if there were differences on which visualization technique to use, we could employ both and then see whether we preferred one over the other in the final product.
- We would probably seek guidance from an instructor or TA if we were unsure of all our options for a decision on analytical methods etc. If we needed help with a starting point it would be best to seek some help earlier on. Also, if we can not come to a consensus on a decision and it would be futile to waste time doing both potential options, we might employ an instructor or TA to be the tie breaker in a sense.

8. Goals and Objectives

What would a successful project look like? Identify a small set of concrete goals and how you will assess them. Consider whether the dashboard answers your research questions, uses appropriately prepared data, presents accurate and interpretable analyses, and is accessible and easy to navigate. What checks will you use to evaluate

data quality, dashboard functionality, and the clarity of your findings? Distinguish essential deliverables from optional features.

- A successful project would match the below goals
    - Satisfy the main criteria of a polished data manipulation project
        - Assessed through the organization in Rstudio/ Github using appropriate folders and reproducibility of the code for use in a different area of interest
    - Conform to the individual goals and interests of each group member
        - Assessed through each team member, spearheading their responsibilities within the project to match what they would ultimately like to gain from this course.
    - The deliverable could be used for forest/land managers for future reference
        - Assessed through whether deliverable is user friendly, functional, accessible, and can be easily interpretable
- What checks will you use to evaluate data quality, dashboard functionality, and the clarity of your findings?
    - Some checks to evaluate data quality are the completeness (lack of NA) for our study area, the uniqueness (lack of duplicate information), integrity (when was the data last updated?) etc.
    - Some checks to evaluate the dashboard functionality is making sure it is intuitive to someone who is only presented with the final deliverable and not the data utilized. Also, that it is accurate and relevant to our target audience, perhaps using a quantitative part that will change given the user-driven location specification.
    - Some checks to evaluate the clarity of our findings to present graphic, tabular, and map visuals as we progress to help the audience understand how we came to the end product. Also, including clear and reproducible steps for our data manipulation and analytical methods used.
- Distinguish essential deliverables from optional features
    - The essential deliverables are some indicator of the relationship of fast fires and the landscape variable of choice (linear regression, etc.) model with statistical R^2 and the visual scatterplot.
    - The other essential deliverable is a map or geographical visualization with the fast fires layer and the layer with the landscape variables
    - An optional feature is the extent to which we can have a dashboard that would show the quantitative composition of the landscape variable within the fast fires boundaries when the user determined the location of interest within the California boundaries.

9. Schedule and Milestones

Work backward from the assignment deadlines to establish internal milestones for:

- Defining research questions and project scope: October 20th at 4pm
- Assessing data feasibility: October 27th at 4pm
- Preparing and combining datasets: October 27th at 4pm
- Conducting exploratory analysis: November 3rd at 4pm
- Building a dashboard prototype: November 10th at 4pm
- Reviewing and testing the dashboard: November 17th at 4pm
- Preparing the final presentation: November 17th at 4pm

Who is responsible for each milestone, how will you track progress, and when will you revisit the scope if necessary?

- We will work as a team and be jointly responsible for ensuring that each of the above milestones is met. In class on Thursdays, we will decide who is responsible for each task that will contribute to these milestones for that specific week. Each member will complete their tasks between Thursday at 4pm and Tuesday at 4pm. We'll meet on Tuesdays at 4pm, with the understanding that the majority of the work should be completed and that this is our time to combine and review the work prior to submitting.
- We will track progress through the running notes document.
- We will revisit the overall scope on a weekly basis as we discuss the milestones for each week and who will complete each task.